

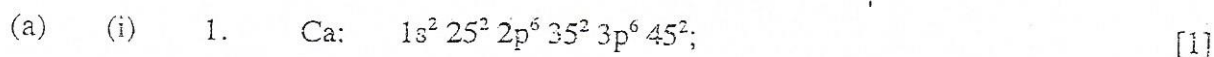
ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

9

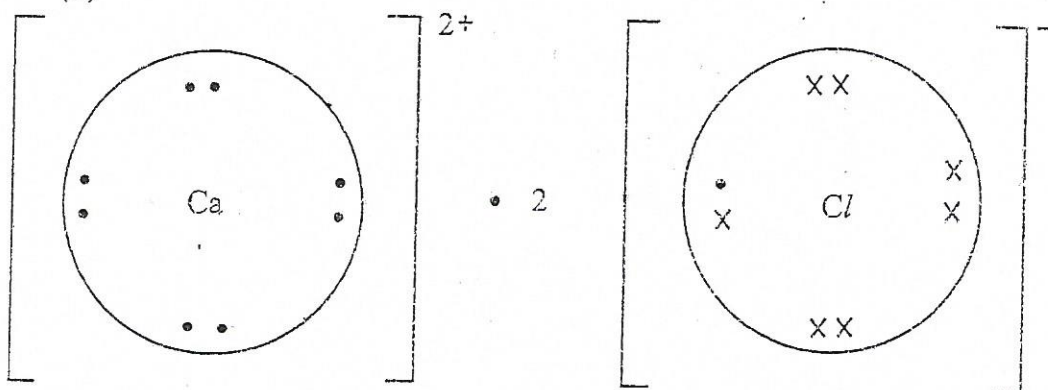
MARKING SCHEME

JUNE 2014

CHEMISTRY 9189/1

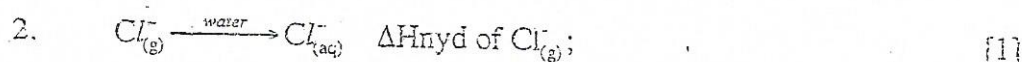
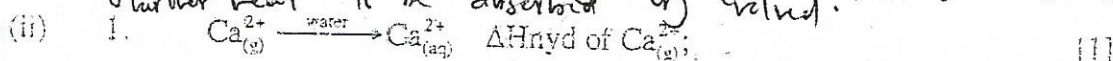


(ii)

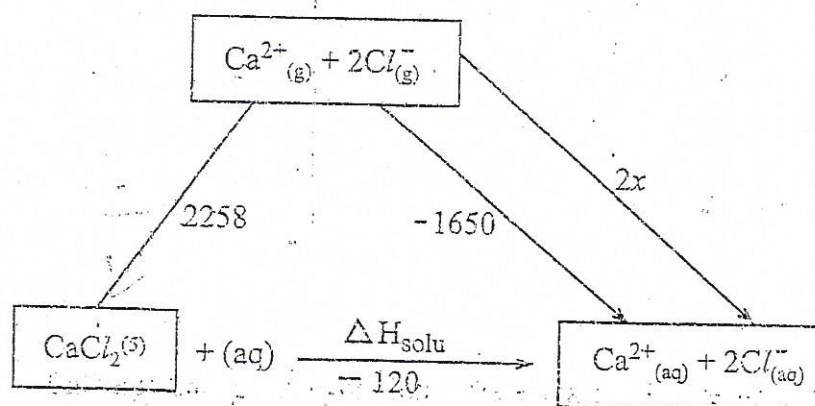


[1]

(b) (i) Enthalpy change of hydration – the enthalpy change when one mole of gaseous ions dissolve in sufficient water to give an infinitely dilute solution; /A.W. *infinitely dilute solution is one where there is a sufficient range excess of H₂O that adding any more doesn't cause any further heat to be absorbed or released.* [1]



(iii) 1.



2. $-120 = 2 \times 258 - 1650 + 2x$; [1]

$2x = -758 - 258$

$\therefore x = -364 \text{ kJmol}^{-1}$; [1]

- (c) (i) Total ΔH_{hyd} of the ions Mg^{2+} and 2Cl^-

$$= -1921 + 2(-364)$$

$$= -2649 \text{ kJmol}^{-1};$$

[1]

- (ii) Mg^{2+} ion smaller than Ca^{2+} ion with larger charge density;

[1]

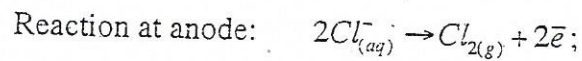
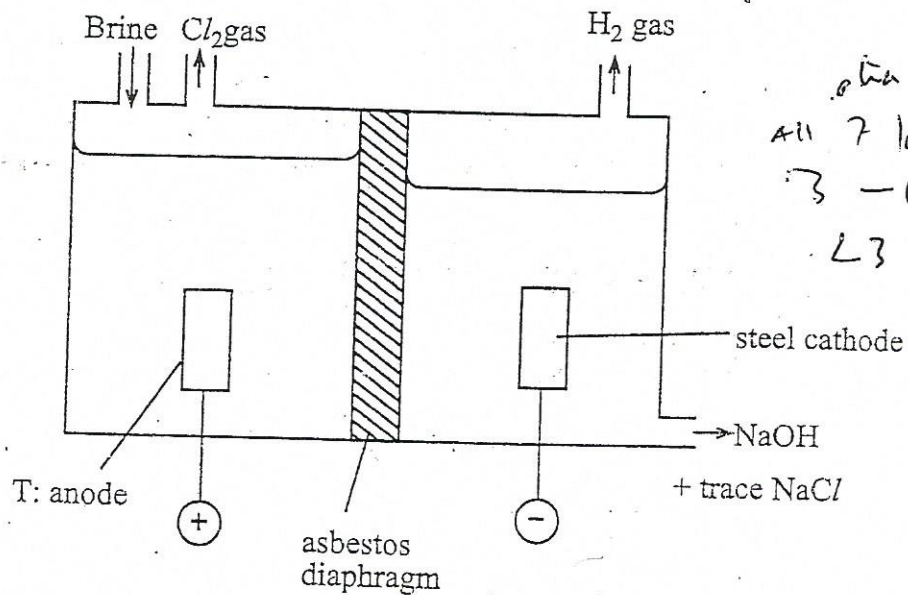
resulting in stronger attractions/bonds with water molecules;

[1]

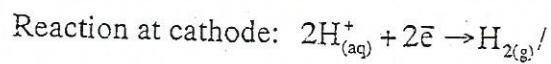
[Total: 12]

2

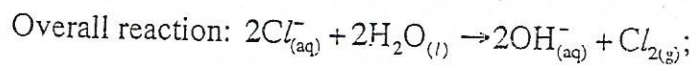
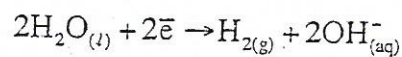
(a)



[1]



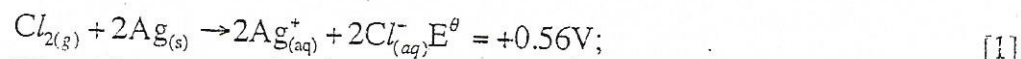
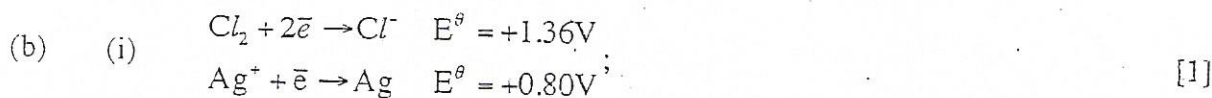
[1]



[1]

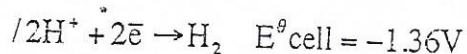
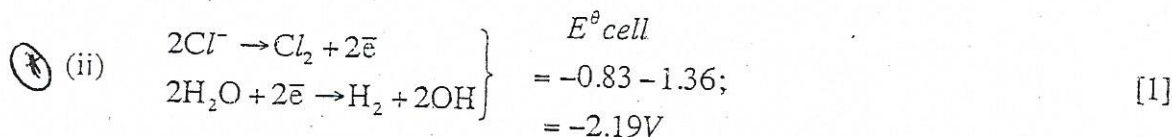
Level of brine in the anode compartment higher than in the cathode compartment to prevent backflow of OH^- which would react with Cl_2 ;

[1]



Cl_2 oxidises Ag to Ag^+ due to $E^\theta_{\text{cell}} > 0$; [1]

$\text{Ag}_{(aq)}^+$ reacts with $\text{Cl}_{(aq)}^-$ ions to form a ppt



Reaction is nonspontaneous since $E^\theta_{\text{cell}} < 0$; hence electricity must be supplied/AW

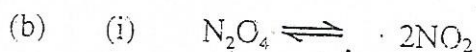
[1]
[Total = 12]

3 (a) (i) LCP – when a constraint is imposed on a system in equilibrium, the system responds in such a way as to oppose the constraint; *Briggs pg 189.* *AW* [1]

(ii) 1. The equilibrium shifts to the left where there are fewer moles of gas; [1]
 colour changes from brown to light brown/pale yellow *photo was* [1]

2. The equilibrium shifts to the right since the reaction is endothermic; [1]

the brown colour becomes more intense/darker; [1]

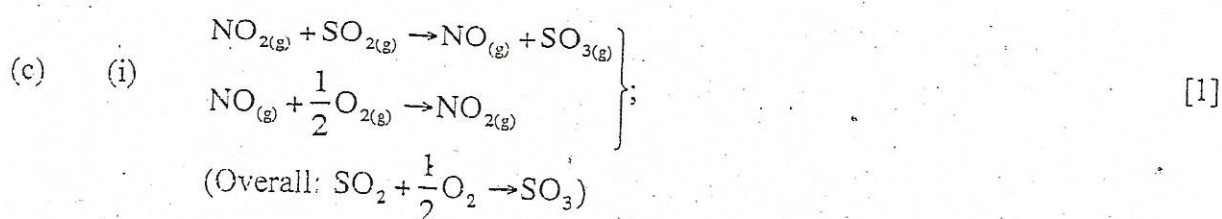


Initial 0.075 0.025
 eqm 0.00275

$\Rightarrow [\text{NO}_2]_{\text{eqm}} = 2(0.075 - 0.00275) + 0.025$

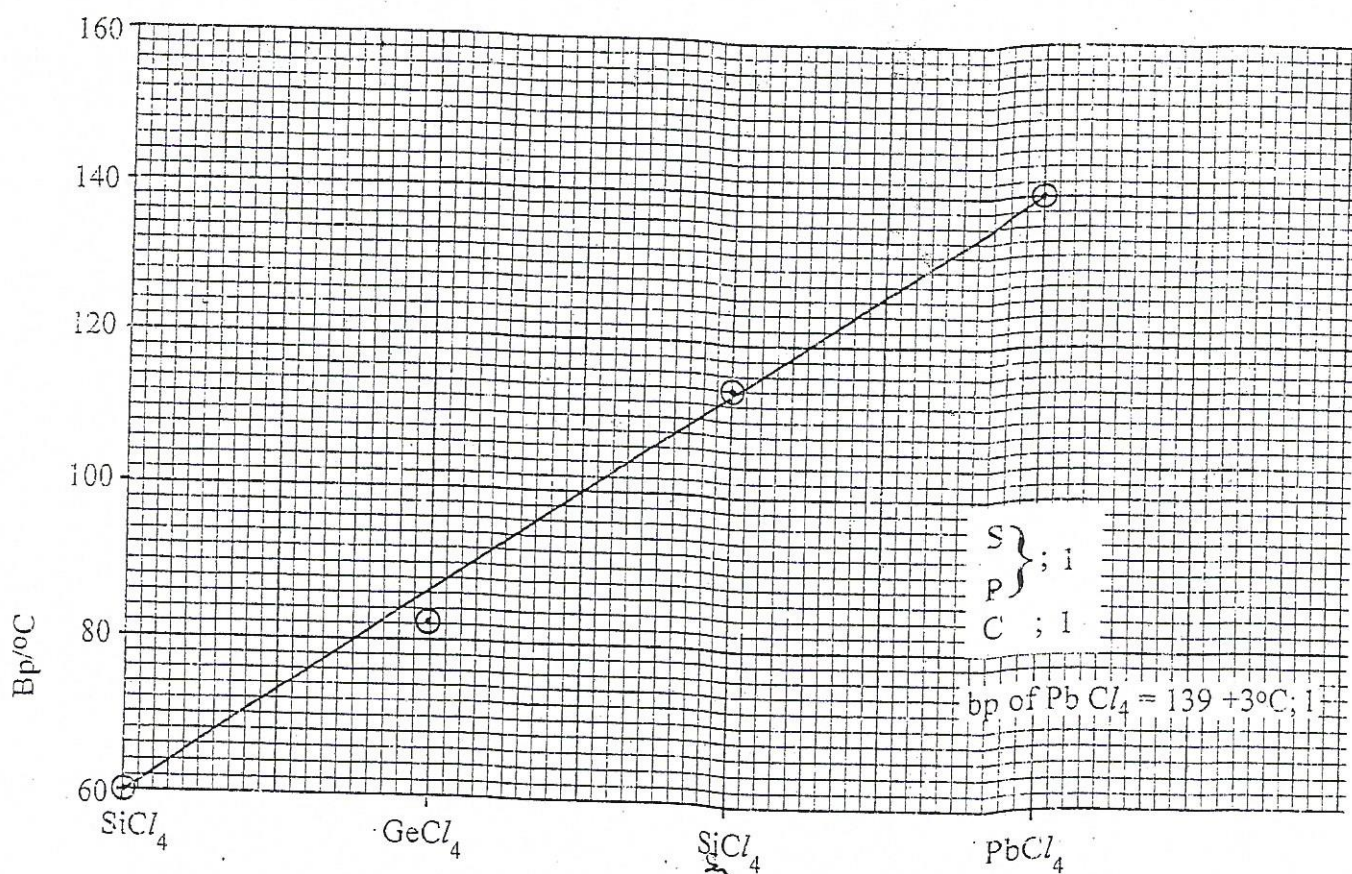
$= 0.1695 \text{ mol dm}^{-3}$; [1]

(ii) $K_c = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} = \frac{0.1695^2}{0.00275} = 10.45 \text{ mol dm}^{-3}$ [3]



- (ii) NO_2 acts as a homogenous catalyst; [1]
 it is in the same phase (gaseous state) with SO_2 and SO_3 ; [1]
 [Total: 12]

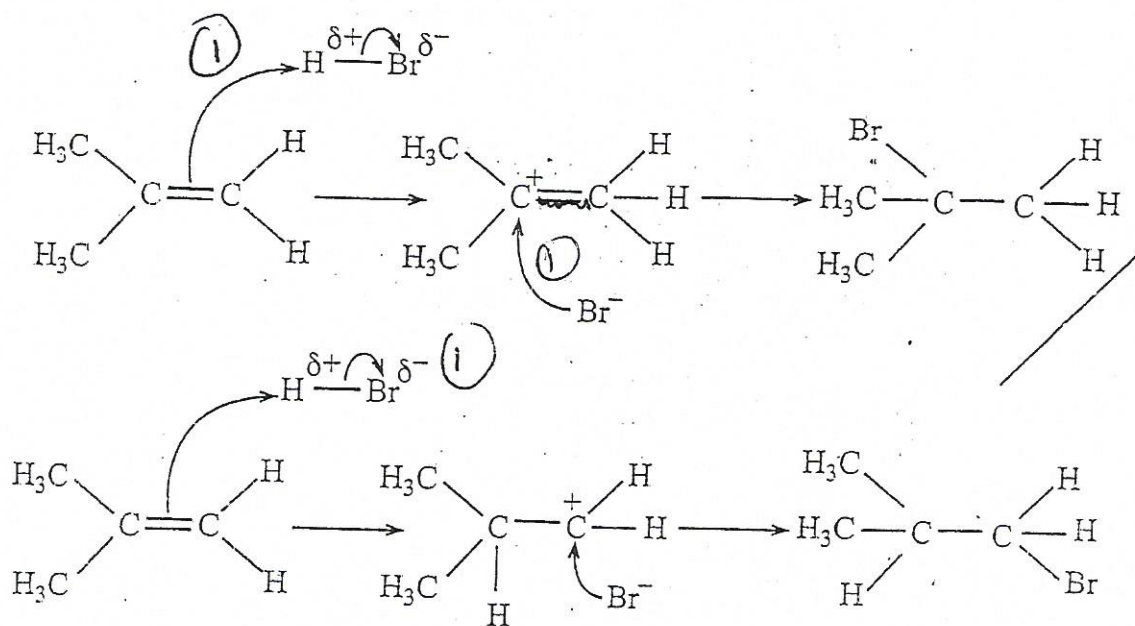
- 4 (a) (i) On graph: appropriate scale } [1]
 points } [1]
 curve; [1]
- (ii) $139^\circ\text{C} \pm 3$; [1]



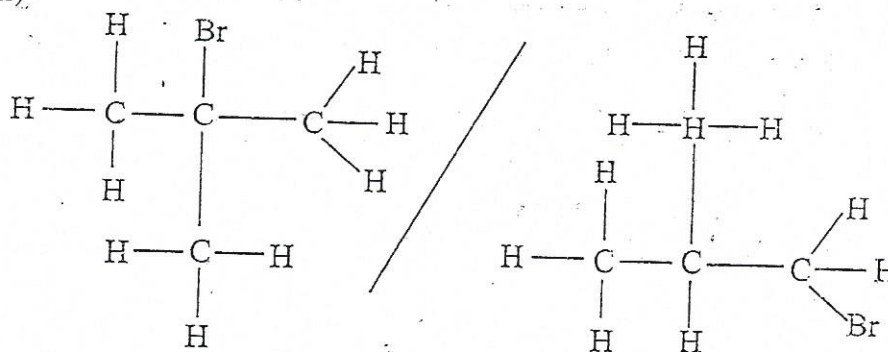
- (iii) PbCl_4 decomposes on heating; [1]

- (a) (i) Electrophilic addition;
electron rich $C=C$ double bond attacked by an electron deficient species; [1]

(ii)



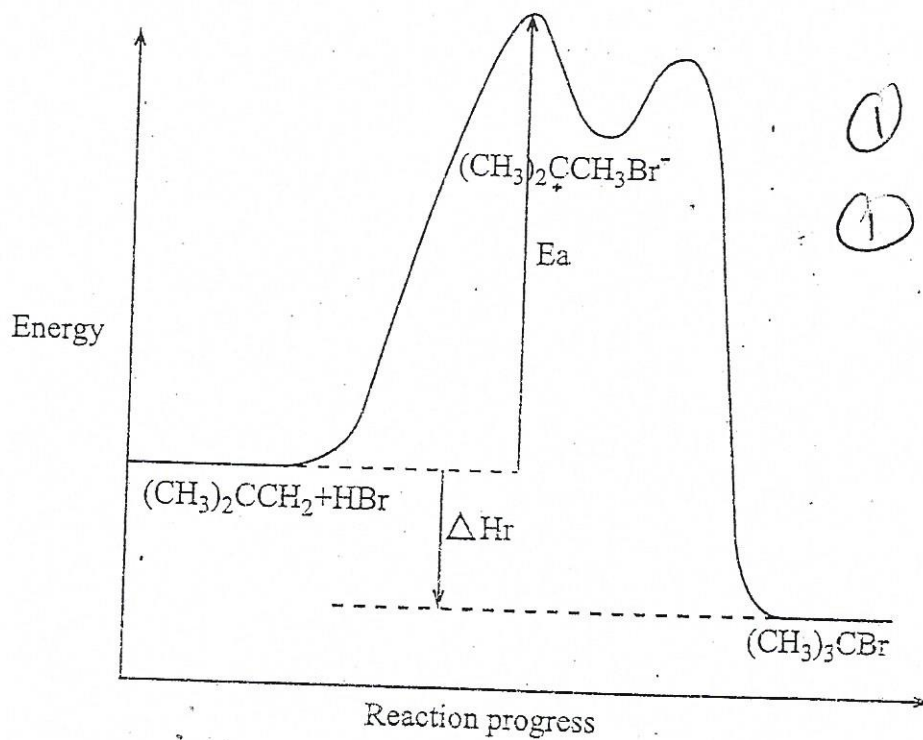
(iii)



- (iv) 2-bromo-2methylpropene/1-bromo-2 methylpropene

[1]

(v)



①

change

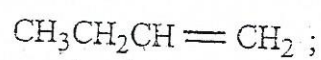
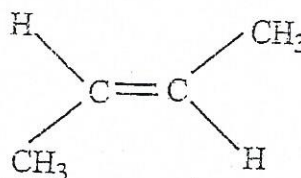
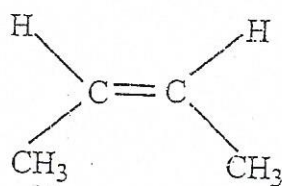
①

labels

(b) (i) Elimination;

[1]

(ii)



Any two

[Total: 12]

(a) (i) $\text{C}_4\text{H}_{10}\text{O}$;

[1]

(ii) Functional group isomerism;

[1]

(iii) warm both with $\text{I}_{2(\text{aq})} + \text{NaOH}$;

[1]

A - no ppt;

[1]

B - yellow ppt/

[1]

Warm with acidified K_2CrO_7 ;

A - didromate (VI) changes from orange to green;

B - no reaction/

Warm with Tollen's/Fehling's reagents;

A - silver mirror/changes from blue to brick red ppt;

B - no reaction;

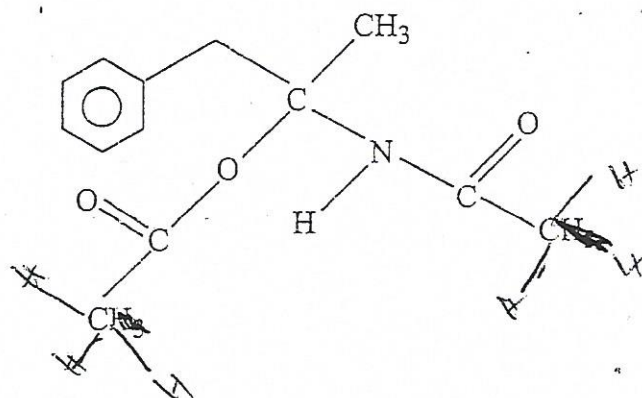
- (b) (i) 1. Nucleophilic addition; [1]
2. Eliminations/dehydration; [1]

- (ii) $\text{NaBH}_4/\text{LiAlH}_4$ at r.t.p./room temperature; [1]
 H_2 , Ni catalysts at 150°C ;

- (c) (i) r.t.p./room temperature; [1]

- (ii) white fumes (of HCl); [1]

(iii)



- (iv) ester/amide; [1]

[Total: 12]

- 8 (a) (i) Polymer – a macromolecule/giant molecule/large molecule made up of many small units (monomers) joined together; AW [1]

- (ii) monomer – a small molecule from which a polymer is made; /AW [1]

- (iii) polymerisation – the process by which many small molecules (called monomers) join together to form a very large molecule; /AW [1]

- (b) (i) 1. condensation; [1]

- (ii) 2. addition; [1]

(w) Amide

- (ii) 1. $\text{HO}_2\text{CCO}_2\text{H}; \text{ClCOCOC l}$ [1]

- $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$; [1]